Employee Performance Management System

Report

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GitHub: https://github.com/XCubee/Employee-performance-management-system/tree/v3

(no config .ini api keys in GitHub)

Problem Statement

Traditional employee performance management systems often rely on yearly reviews, lack regular feedback and do not provide enough transparency in the evaluation process. Employees may find it difficult to track their progress, understand their scores or raise concerns about unfair evaluations. Managers may also struggle to monitor employee workloads and assign tasks effectively. Therefore, there is a need for an Employee Performance Management System (EPMS) that makes performance tracking easier, improves communication between employees and managers, and ensures a fair, transparent and continuous review process.

1. About the Project

EPMS is a web application developed to track employee performance and manage their work. It has three types of logins: HR, Manager, and Employee, and each one has a separate dashboard. Managers can assign goals and tasks to employees, review their performance, and track their progress. Employees can check their marks and performance details and also raise an appeal if they feel that something is unfair. The main aim of this project is to make the performance review process more transparent and regular instead of depending only on yearly reviews.

2. What We Added to the project

Before starting the development, we first went through some research papers related to performance management systems to understand how an effective PMS should work and what features are important. Based on our research, we focused mainly on fairness, regular feedback, clear goal setting and employee involvement in the performance review process.

In the first two versions, we developed the basic features of the system, such as login, goal assignment, performance reviews and feedback. As we moved to Version 3, we added more features to make the system better and more useful. These included employee profiles, manager approval for employee progress updates, a three-part marking system, a separate metrics page and an appeal system where employees can raise concerns if they feel their evaluation is unfair.

We also improved the task assignment process by adding employee workload, skills and experience details so that managers can assign tasks more effectively based on each employee's ability and current workload. For developing and managing the project, we used technologies like Flask, Supabase, HTML, Jinja, Git and GitHub. Overall, we developed the project step by step, adding new features and improvements in each version based on our requirements.

Steps to Run

1. Unzip the zip file
2. pip install –r requirements.txt
3. Initialize venv
4. Python app.py

Note – The github doesn't contain config.ini and does which contains the API keys

The zip file contains the API keys so pls unzip and use that for full functionality

Login credentials

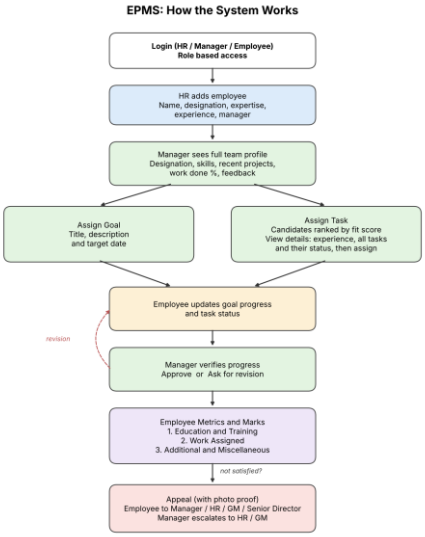
Hr – sakshamahluwalia2802@gmail.com password – saksham

Manager-- sakshamahluwalia@gmail.com password – saksham

Employee – test@gmail.com password – test

STACK – HTML + PYTHON +SUPABASE(DB) for modularity in any device

3. Process Diagram



The flow is simple. HR adds an employee. The manager gives goals and tasks. The employee updates the work. The manager checks it and either approves it or sends it back. After that the marks and metrics show up for the employee, and if they are not happy with it, they can file an appeal.

4. Project Screenshots

Below are screenshots of the main pages, with a short note on what each page does.

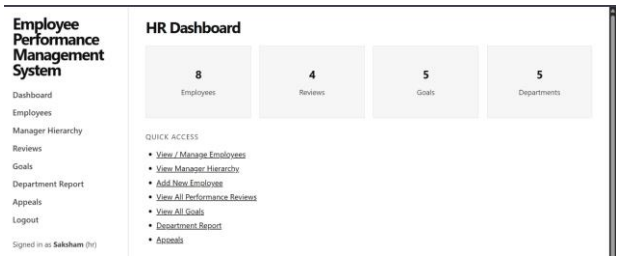


Figure 1: HR Dashboard

This is the first page HR sees after login. The cards on top show how many employees, reviews, goals and departments are there in the system. Below that, HR gets quick links to add employees, see the hierarchy, check reviews and handle appeals.



Figure 2: Employee Hierarchy

Here HR can see who reports to whom. Every employee is shown under their manager with their name and designation, so the structure of the team is clear in one look.



Figure 3: All Performance Reviews

This page lists every review given in the company. For each one, HR can see the employee, the person who reviewed them, the period, the rating out of 5 and the comments.

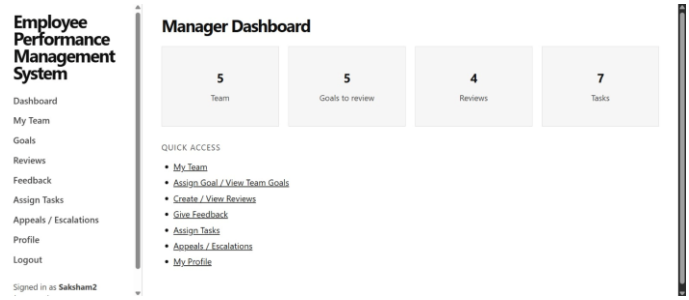


Figure 4: Manager Dashboard

When a manager logs in, they see their team size, goals waiting for review, reviews written and tasks given. The quick links take them to the main things they do daily.

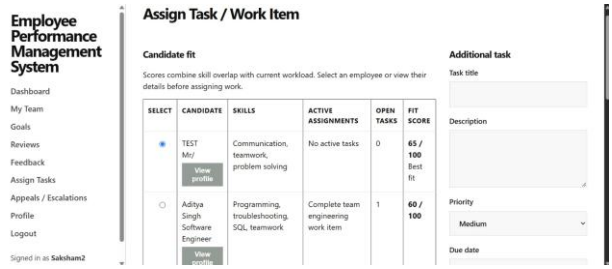


Figure 5: Assign Task

Before giving a task, the manager sees a list of team members with their skills, open tasks and a fit score out of 100. The person with the best fit comes on top. The "View profile" button shows more details about the person, and the form on the right is used to assign the task.

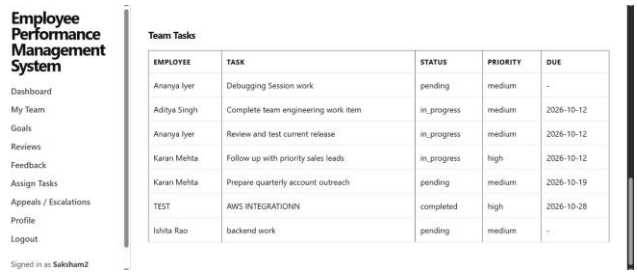


Figure 6: Team Tasks

This table shows all tasks given to the team. The manager can check who is doing what, the current status, priority and due date of each task.

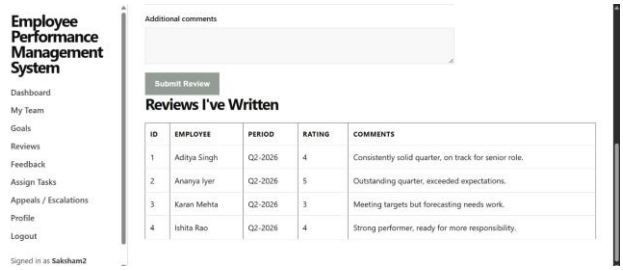


Figure 7: Writing Reviews

The manager fills a short form with rating and comments to review a team member. All reviews written by the manager are listed below the form.

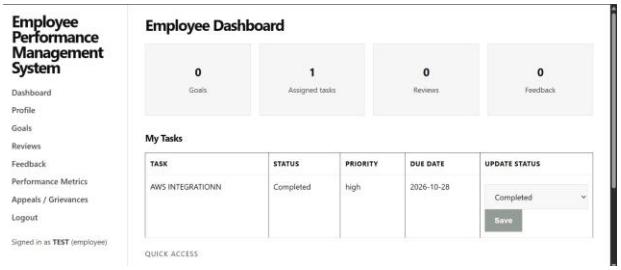


Figure 8: Employee Dashboard

The employee sees how many goals, tasks, reviews and feedback they have. Under "My Tasks" they can change the status of a task and click Save, so the manager knows the progress.

5. Summary and Result

We were able to develop a working EPMS where HR, managers and employees have their own separate access and dashboards. In Version 3, most of our focus was on making the system fairer and more transparent for employees by adding features that allow them to understand their performance and raise concerns if they feel something is not right. The table below shows how we have included the important points from the research papers in our project.

All the features and changes that we planned for Version 3 have been completed and successfully pushed to the v3 branch on GitHub

6. Final Outcome

We have successfully developed a working web-based EPMS with separate access for HR, managers and employees. Managers can check an employee's profile, skills and current workload before assigning them new tasks. Employees can view their marks, assigned tasks, goals and feedback all in one place, which makes it easier for them to track their performance. Also, employee progress is only counted after it is approved by the manager, and employees have a proper appeal system where they can raise complaints if they feel their evaluation is unfair.

7. Future Work

- Add AI to help managers write feedback and to suggest training, while keeping employee data private and the results unbiased.
- Calculate marks fully from real data. Right now some default marks are estimated when HR has not entered them.
- Add notifications, feedback from teammates, self appraisal and charts that show progress over time.
- Make the site work better on mobile and switch to a stronger password hashing method like bcrypt.

8. References

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2. Rahmita, S., Fahmy, R., & Syahrul, L. (2025). The building blocks of an effective performance management system: A comprehensive review. Journal of Management and Digital Business, 5(1), 13-34.. https://doi.org/10.53088/jmdb.v5i1.1512

3. Duda, J., Flynn, J., Cantrell, S., Scoble-Williams, N., & Sanford, A. (2025, March 23). Reinventing performance management processes won't unlock human performance. Here's what will. Deloitte Insights.

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